Hossein Kavianirad

Doctoral Researcher, Information-oriented Control (ITR), Technical University of Munich

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Research Summary

My research interests include control theory, learning-based control, human-centric control, and shared and cooperative control of hybrid exoskeletons and wearable robots. I have hands-on experience in robot design and control and in hardware-in-the-loop experiments. During my PhD, I have developed controllers for input-redundant systems (e.g., hybrid exoskeletons) that distribute the control effort between actuators, such as an exoskeleton and functional electrical stimulation (FES), while taking into account the user's intent. I have used dynamic allocation with stability guarantees, model predictive control (MPC), and EMG-based intent estimation, and have tested these methods with healthy participants and stroke patients on several robotic platforms. In my M.Sc., I worked on Lyapunov-based boundary control of systems described by partial differential equations (PDEs).

Education

04/2021–03/2027 (expected) **Doctoral candidate, Electrical Engineering** — Technical University of Munich
Chair of Information-oriented Control (ITR), TUM School of CIT. Supervisor: Prof. Dr.-Ing. Sandra Hirche.
Dissertation: *Cooperative Shared Control of Hybrid FES–Exoskeleton.*

09/2015–09/2017 **M.Sc. Mechanical Engineering (Applied Mechanics)** — Sharif University of Technology, Tehran
GPA 19.27/20, ranked 2nd of 151. Thesis: *Boundary Control of Vibration in Satellite Flexible Solar Panels* (Lyapunov-based boundary control, asymptotic and exponential stability, disturbance observer). Supervisors: Prof. H. Salarieh, Prof. A. Alasty.

09/2011–09/2015 **B.Sc. Mechanical Engineering** — Shiraz University
GPA 19.05/20, ranked 1st of 89. Project: *Energy harvesting in a piezoelastic beam excited by air flow* (supervisor: Prof. M. Farid).

Research Experience

04/2021–present **Doctoral Researcher** — ITR, Technical University of Munich

- Designed shared and cooperative control for hybrid FES–exoskeletons: impedance-based shared control, MPC-based FES control, and a real-time dynamic allocator that redistributes the required torque between FES and exoskeleton under heterogeneous actuator constraints, with a stability proof and analysis. Extended the allocator to personalized allocation based on participant-specific attainable sets.
- Identified user-specific models (nonlinear, delayed, time-varying neuromuscular dynamics) from sparse data using learning-based methods, and estimated volitional torque from EMG using a novel adaptive filter [Link].
- Conducted studies with healthy participants and stroke patients on the NESM-γ elbow exoskeleton (Scuola Superiore Sant'Anna), the HRX-1 wrist robot (HumanRobotiX), and FES/EMG hardware (Tecnalia) within the EU Horizon 2020 project ReHyb (GA 871767), in collaboration with Imperial College London and Schön Klinik Bad Aibling.
- Developing a hand prosthesis driven by high-density EMG for dexterous hand movements (ongoing).

10/2017–03/2021 **Researcher** — Shiraz University Research Center
Worked on system identification and control; programmed embedded ARM control systems (Keil) with sensors and actuators in C/C++.

Honors and Awards

- Best Paper Award, IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES), 2022.
- Ranked 2nd of 151 (M.Sc., Sharif University of Technology); ranked 1st of 89 (B.Sc., Shiraz University).
- Iran's National Elites Foundation: fellowship (2014–2015) and funded teaching-assistant position (2015–2017).
- Distinguished student of the mechanical engineering department, Sharif University of Technology (2015–2016) and Shiraz University (2014–2015).

Teaching and Supervision

- **Advanced Nonlinear Control** (M.Sc. course, ITR, TUM), 2021–present: supervision of student tutorial reports on advanced nonlinear-control topics, reviews, and presentations (five semesters).
- **Advanced Control and Robotics Lab** (M.Sc. lab, LSR and ITR, TUM), SS 2023–SS 2026: supervisor of the *Sliding Mode Control* experiment (four semesters).
- **Thesis and project supervision, TUM:** 16 students (7 B.Sc. theses, 1 M.Sc. thesis, 2 research internships, and 6 engineering internships).
- **Sharif University of Technology (M.Sc.):** teaching assistant for Advanced Control (M.Sc., Fall 2016)

and Dynamics (Fall 2015); laboratory assistant, Dynamics of Machinery and Vibrations Lab (three semesters, 2015–2016).

Technical Skills

Programming	MATLAB/Simulink, C/C++, embedded ARM programming (Keil), Python, Git/GitHub, Claude Code
Simulation and CAD	MuJoCo, Simulink, Abaqus, SolidWorks
Hardware	Exoskeletons (elbow, wrist), soft exoglove, prosthetic and dexterous robotic hands, multi-array FES electrodes, EMG and high-density EMG (HD-EMG), force/torque sensors, IMUs, Qualisys motion capture, embedded real-time controllers
Languages	English (fluent), German (basic knowledge), Persian (native)

Publications

Selected publications

[1] **H. Kavianirad**, S. Endo, D. Astarita, L. Amato, E. Trigili, S. Hirche, "Cooperative control of hybrid FES-exoskeleton: Dynamic allocation," under review at *IEEE Transactions on Control Systems Technology*; arXiv preprint, 2025. doi:10.48550/arXiv.2511.10117

[2] **H. Kavianirad**, F. Missiroli, S. Endo, L. Masia, S. Hirche, "Toward dexterous hand functional movement: Wearable hybrid soft exoglove-FES study," *IEEE BioRob*, pp. 1346–1351, 2024. doi:10.1109/BioRob60516.2024.10719731

[3] **H. Kavianirad**, M. Forouhar, H. Sadeghian, S. Endo, S. Haddadin, S. Hirche, "Model-based shared control of a hybrid FES-exoskeleton: An application in participant-specific robotic rehabilitation," *Int. Conf. Rehabilitation Robotics (ICORR)*, 2023. doi:10.1109/ICORR58425.2023.10304764

[4] **H. Kavianirad**, S. Endo, T. Keller, S. Hirche, "EMG-based volitional torque estimation in functional electrical stimulation control," *IEEE-EMBS IECBES*, pp. 171–176, 2022. **Best Paper Award.** doi:10.1109/IECBES54088.2022.10079376

[5] **H. Kavianirad**, H. Salarieh, A. Alasty, R. Vatankhah, "Boundary control of flexible satellite vibration in planar motion," *Journal of Sound and Vibration*, vol. 432, pp. 549–568, 2018. doi:10.1016/j.jsv.2018.06.052

[6] **H. Kavianirad**, H. Salarieh, A. Alasty, R. Vatankhah, "Boundary control of anti-symmetric vibration of satellite with flexible appendages in planar motion with exponential stability," *Acta Astronautica*, vol. 147, pp. 219–230, 2018. doi:10.1016/j.actaastro.2018.03.050

[7] **H. Kavianirad**, H. Nejat Pishkenari, "Frictional viscoelastic based model for spherical particles collision," *Granular Matter*, vol. 20, no. 4, art. 62, 2018. doi:10.1007/s10035-018-0835-9

Co-authored publications

[8] L. Cazenave, C. Höhler, S. Endo, **H. Kavianirad**, N. Peña-Pérez, L.-M. Huber, A. Yurkewich, T. Keller, S. Hirche, K. Jahn, et al., "Hybrid robot & FES rehabilitation after stroke: Short-term performance, fatigue, and patient experience," *J. Rehabil. Assist. Technol. Eng.*, vol. 13, 2026. doi:10.1177/20556683261488467

[9] C. Höhler, S. Endo, J. Hermsdörfer, L. Cazenave, **H. Kavianirad**, K. Jahn, C. Krewer, "Repetitive grasping after stroke assisted by functional electrical stimulation," *Artificial Organs*, 2025. doi:10.1111/aor.15033

[10] N. Das, S. Endo, **H. Kavianirad**, S. Hirche, "Framework for learning a hand intent recognition model from sEMG for FES-based control," *IEEE BioRob*, pp. 1320–1327, 2024. doi:10.1109/BioRob60516.2024.10719910

[11] A. Sena, **H. Kavianirad**, S. Endo, E. Burdet, S. Hirche, "The gap in functional electrical stimulation simulation," *3rd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics*, 2022. Full text (mediaTUM)

[12] H. Nejat Pishkenari, **H. Kavianirad**, H. Jafari Shad, "Transformation of sliding motion to rolling during spheres collision," *Granular Matter*, vol. 19, no. 4, art. 70, 2017. doi:10.1007/s10035-017-0755-0

References

Available on request.